Artificial Intelligence

BS (CS) Spring 2026

Supervised Learning

Learning Objectives:

1. Supervised Learning
2. Linear Regression
3. Logistic Regression
4. KNN

Lab Tasks

Submission Instructions

1. Create one notebook file for all questions with name *ROLLNO_SEC_LAB12* **e.g. i23XXXX_A_LAB012**
2. Submit your .ipynb file with correct name on Google Classroom under your section heading.
3. If you don't follow the above-mentioned submission instruction, you will be marked **zero.**

Plagiarism in the Lab Task will result in **zero** marks in the whole category.

Questions

Q1. You are provided with a dataset containing information about car purchasing, including features such as area, number of rooms, location, price, and other relevant attributes.

Your task is to:

1. Perform **Exploratory Data Analysis (EDA)** on the dataset. This should include data cleaning, handling missing values, outlier detection, and feature insight analysis.
2. Conduct **statistical analysis** of the dataset. This may include distribution analysis, correlation analysis, variance checks, skewness, kurtosis, and hypothesis testing where applicable.
3. Preprocess the data by encoding categorical variables and scaling numerical features if required.
4. Split the dataset into training and testing sets.
5. Train an appropriate **machine learning model** to predict property prices.
6. Evaluate the model using suitable performance metrics.
7. Make predictions on the test data.

Finally, **interpret the predictions and results**, explaining how different features impact the predicted real estate prices.